Session – 12

Task-1.

Create a CTE using the WITH clause to select all products with a rating above 4.5 from a 'Products' table, similar to how Flipkart or Myntra might highlight top-rated items.

Solution:

```
WITH TopRatedProducts AS (
  SELECT *
  FROM Products
  WHERE rating > 4.5
)
SELECT *
FROM TopRatedProducts;
```

Task-2.

Rewrite a query that finds all restaurants in 'Ahmedabad' with delivery charges under 50 from a 'Restaurants' table, first using a subquery and then using a CTE. Compare both queries for readability.

Hint: Focus on making the CTE version cleaner and easier to understand.

Solution:

```
SELECT *
FROM (
  SELECT *
  FROM Restaurants
  WHERE city = 'Ahmedabad'
    AND delivery_charge < 50
) AS AffordableRestaurants;
```

```
WITH AffordableRestaurants AS (
  SELECT *
  FROM Restaurants
  WHERE city = 'Ahmedabad'
    AND delivery_charge < 50
```

```
)
SELECT *
FROM AffordableRestaurants;
```

Task-3.

Using two CTEs in a single query, find the top 3 most-followed users and the top 3 most-liked posts from a 'Users' and 'Posts' table (think Instagram-style data). Output both lists in the same result set.

Solution:

```
WITH TopUsers AS (
  SELECT
    id,
    username,
    followers
  FROM Users
  ORDER BY followers DESC
  LIMIT 3
),

TopPosts AS (
  SELECT
    id,
    content,
    likes
  FROM Posts
  ORDER BY likes DESC
  LIMIT 3
)

SELECT
  'User' AS type,
  username AS name,
```

```
  followers AS count
FROM TopUsers

UNION ALL

SELECT
  'Post' AS type,
  content AS name,
  likes AS count
FROM TopPosts;
```

Task-4.

Write a recursive CTE that generates a list of dates for the next 7 days starting from today, similar to how BookMyShow shows available dates for movie bookings.

Hint: Use a base case for today and recursion to add one day at a time.

Solution:

```
WITH RECURSIVE Dates AS (
  -- Base case
  SELECT CURRENT_DATE AS booking_date

  UNION ALL

  -- Recursive case
  SELECT booking_date + INTERVAL 1 DAY
  FROM Dates
  WHERE booking_date < CURRENT_DATE + INTERVAL 6 DAY
)

SELECT booking_date
FROM Dates;
```

Task-5.

Given a messy SQL query that finds all users with more than 1000 followers from a 'Users' table, refactor it to use a CTE for better clarity and maintainability.

Solution:

```
WITH PopularUsers AS (
  SELECT
    id,
    username,
    followers
  FROM Users
  WHERE followers > 1000
)
SELECT *
FROM PopularUsers;
```